Non-collapse of median-normalized shortest directed paths

A marked-quantum-wedge regeneration argument

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Abstract

Let M be the uniform infinite bipolar-oriented triangulation (UIBOT). Give an edge its north-going and west-going Peano times, divide both times by n, and divide shortest directed-path (SDP) distance by a median m_n of the maximal finite SDP distance in an n-increment KMSW cell. Assume that along a deterministic subsequence the resulting closed distance graphs converge, jointly with the primal/dual peanosphere data, to the graph of a finite-valued function D. We prove that D is almost surely not identically zero.

The seed is discrete and local: with probability bounded away from zero, an n-increment cell contains an SDP geodesic of length in $[m_n, Am_n]$, and a two-buffer KMSW locality argument makes the middle of this geodesic survive arbitrary sewing outside the cell. The probability-one amplification is then continuum. We extract countably many independent marks on disjoint unit quantum-area cells of the peanosphere. Gwynne–Holden–Sun identify each unmarked cell with an intrinsic piece of a fresh weight-4/3 quantum wedge. The marks may retain microscopic randomness invisible to the Brownian peanosphere, but remain independent because independence is inherited before taking the weak limit. A product-space Borel–Cantelli argument gives a successful continuum cell almost surely. The joint primal/dual estimates of Gwynne–Holden–Sun prevent the adaptively chosen witness from escaping in the transverse Peano coordinate, so success yields a positive-height point of the limiting SDP graph.

No comparison between $m_{\lfloor cn \rfloor}$ and m_n is used. The argument is conditional on graphness of the chosen subsequential limit and does not prove point separation.

1 Setup and statement

Let

$$\lambda^N, \lambda^W : \mathbb{Z} \to E(M)$$

be the north-going and west-going edge-bijective Peano orders, with the deterministic convention of [3]. For $e \in E(M)$ put

$$i(e) = (\lambda^N)^{-1}(e), \qquad j(e) = (\lambda^W)^{-1}(e), \qquad p(e) = (i(e), j(e)) \in \mathbb{Z}^2.$$

The Baxter/BBMF correspondence gives

$$p(e) \le p(f) \quad \Longleftrightarrow \quad \text{there is a directed path whose first edge is } e \text{ and last edge is } f, \tag{1}$$

where the inequality is coordinatewise [1].

For comparable edges let $d^{\bullet}_M(e, f)$ be the smallest number of edges in such a directed path; set $d^{\bullet}_M(e, e) = 1$. Let $M^{\circ}_{0,n}$ be a fresh KMSW cell encoded by n independent triangulation increments and let

$$\Delta_n := \max\{\mathrm{SDP}_{M^{\circ}_{0,n}}(u, v) : \mathrm{SDP}_{M^{\circ}_{0,n}}(u, v) < \infty\}.$$

Choose the integer median

$$m_n := \max\{m \in \mathbb{N} : \mathbb{P}[\Delta_n \geq m] \geq 1/2\}. \tag{2}$$

Thus

$$\mathbb{P}[\Delta_n \geq m_n] \geq \frac{1}{2}, \qquad \mathbb{P}[\Delta_n \geq m_n + 1] < \frac{1}{2}. \tag{3}$$

The finite-volume lower bound of Borga–Gwynne implies $m_n \to \infty$; in fact $m_n = n^{3/8+o(1)}$ [2, Theorem 1.13].

Set

$$\mathcal{U}_n := \{n^{-1}p(e) : e \in E(M)\}, \qquad \mathcal{W}_n := \{(x, y) \in \mathcal{U}_n^2 : x \leq y\}.$$

For $x = n^{-1}p(e)$ and $y = n^{-1}p(f)$ define

$$D_n(x, y) := \frac{d_M^\bullet(e, f)}{m_n}, \qquad \Gamma_n := \mathrm{Graph}(\mathcal{W}_n, D_n) \subset \mathbb{R}^5.$$

We use the Fell topology on closed subsets of Euclidean space. Write Z_n for the usual linearly interpolated diffusive rescaling of the KMSW encoding walk, normalized as in [3]; then Z_n converges to the peanosphere Brownian motion Z.

Input 1.1 (Graph-valued subsequential limit)**.** *Along a deterministic sequence $n_r \to \infty$, jointly with the rescaled primal/dual peanosphere data,*

$$(\mathcal{W}_{n_r}, \Gamma_{n_r}, Z_{n_r}) \Longrightarrow (\mathcal{W}, \Gamma, Z),$$

and almost surely

$$\Gamma = \{(w, D(w)) : w \in \mathcal{W}\}$$

for a finite-valued function $D : \mathcal{W} \to [0, \infty)$.

Theorem 1.2 (Non-collapse)**.** *Under Input 1.1,*

$$\mathbb{P}[\exists\, x \neq y :\ (x, y) \in \mathcal{W},\ D(x, y) > 0] = 1.$$

In the joint realization constructed below, almost surely infinitely many pairwise disjoint unit north-Peano cells contain such a positive witness.

Remark 1.3. *The theorem is deliberately conditional on graphness. The present proof does not establish a median-normalized endpoint modulus. Its purpose is only to show that a graph-valued subsequential limit cannot collapse identically to zero.*

2 A local median-scale seed

2.1 Interval locality and two-buffer persistence

We use two elementary consequences of the KMSW sewing construction. First, north-going active-edge labels strictly increase along every directed path. Second, for an increment interval $(a, b]_{\mathbb{Z}}$, incidences among the active edges $\lambda^N(a), \ldots, \lambda^N(b)$ depend only on the increments in that interval. These are the interval forms of the KMSW endpoint identities [4].

Lemma 2.1 (Two-buffer persistence)**.** *Suppose a fresh increment block contains a vertex SDP geodesic*

$$z_0 \xrightarrow{e_1} z_1 \xrightarrow{e_2} \cdots \xrightarrow{e_d} z_d, \qquad d \geq 3.$$

After arbitrary KMSW increments are sewn before and after the block,

$$\mathrm{SDP}_M(z_1, z_{d-1}) = d-2, \qquad d_M^{\bullet}(e_2, e_{d-1}) = d-2. \tag{4}$$

Proof. The middle segment has $d-2$ edges. If the completed map contained a shorter path Q from z_1 to z_{d-1}, then every active label of Q would lie strictly between the labels of e_1 and e_d, since active labels increase along directed paths. Hence all edges and incidences of Q are internal to the original increment block. Interval locality puts the same Q in the fresh cell. Concatenating e_1, Q, and e_d would shorten the original geodesic, a contradiction. The edge-to-edge statement follows from the same argument. □

2.2 A seed probability bounded away from zero

Lemma 2.2 (Uniform bounded-height seed)**.** *There are universal constants $A < \infty$ and $c_0 > 0$ such that for every sufficiently large n,*

$$\mathbb{P}[m_n \leq \Delta_n \leq A m_n] \geq c_0. \tag{5}$$

One may take $A = 16$ and $c_0 = 7/16$.

Proof. Let $p_n(r) = \mathbb{P}[\Delta_n \geq r]$. For $r, k \in \mathbb{N}$,

$$p_n(k(r+2)) \leq p_n(r)^k. \tag{6}$$

Indeed, if a geodesic has at least $k(r+2)$ edges, divide its first $k(r+2)$ edges into k consecutive packets of $r+2$ edges. The increment intervals strictly between successive packet endpoint labels are pairwise disjoint. By Lemma 2.1, fixing the increments in one such interval forces an SDP distance at least r regardless of all increments outside that interval. Thus the k intervals witness k disjoint occurrences of $\{\Delta_n \geq r\}$. Reimer's inequality gives (6) [5].

Apply (6) with $r = m_n + 1$ and $k = 4$. By (3),

$$\mathbb{P}[\Delta_n \geq 4(m_n+3)] \leq 2^{-4}.$$

Since $m_n \geq 1$, $4(m_n+3) \leq 16 m_n$, so

$$\mathbb{P}[\Delta_n > 16 m_n] \leq 2^{-4}.$$

Together with $\mathbb{P}[\Delta_n \geq m_n] \geq 1/2$, this proves (5) with the stated constants. □

3 Uniform control of the transverse Peano coordinate

A successful cell supplies an adaptively chosen SDP maximizer. Therefore convergence for a fixed active edge is insufficient: we need tightness of the west-going labels of *all* active edges in one deterministic north-going block.

For integers $a < b$, GHS Section 5.1 defines north and south pocket edge sets $M^N_{a,b}$ and $M^S_{a,b}$ using the north- and south-going dual flow lines started from $\lambda^N(a)$ and $\lambda^N(b)$. GHS Proposition 5.1 states that for fixed real $s < t$,

$$\left\{ n^{-1} \# M^N_{\lfloor sn \rfloor, \lfloor tn \rfloor} \right\}_{n \geq 1}, \qquad \left\{ n^{-1} \# M^S_{\lfloor sn \rfloor, \lfloor tn \rfloor} \right\}_{n \geq 1} \tag{7}$$

are tight [3].

Lemma 3.1 (Deterministic flow-line enclosure)**.** *Let $a \le r \le b$. Let $\mathcal{R}_{a,r}$ be the finite region disconnected from infinity by the north- and south-going dual flow lines from $\lambda^N(a)$ and $\lambda^N(r)$. Then, up to a universal number of boundary edges,*

$$\mathcal{R}_{a,r} \subset \lambda^N([a,b]_{\mathbb{Z}}) \cup M^N_{a,b} \cup M^S_{a,b}. \tag{8}$$

Proof. Branches of each discrete flow-line tree do not cross and merge after their first intersection. Hence the north-going branch from an intermediate edge $\lambda^N(r)$ cannot leave the planar strip between the two endpoint north-going branches without first meeting one of them; after such a meeting it coincides with that branch. The same statement holds for the south-going branches. Consequently, the part of the four-flow-line region $\mathcal{R}_{a,r}$ which lies after the north-going exploration has left the active slab $[a,b]_{\mathbb{Z}}$ is trapped in the endpoint north pocket $M^N_{a,b}$, and the part lying before the exploration enters the slab is trapped in the endpoint south pocket $M^S_{a,b}$. All remaining edges are among those swept by λ^N between active times a and b. This is exactly (8). The endpoint conventions in the augmented map change the count by at most a universal constant. □

Lemma 3.2 (Block transverse compactness)**.** *Fix rational $s < t$ and put $a_n = \lfloor sn \rfloor$ and $b_n = \lfloor tn \rfloor$. Then*

$$\left\{ n^{-1} \max_{a_n \le r \le b_n} |j(\lambda^N(r))| \right\}_{n \ge 1} \tag{9}$$

is tight. The statement holds simultaneously for any fixed finite family of rational macroscopic intervals.

Proof. Let $T_n(r) = j(\lambda^N(r))$. GHS Lemma 2.5 identifies this, up to the fixed orientation/time convention, with the dual west-going Peano time of the edge crossing $\lambda^N(r)$. The observation immediately following GHS Proposition 5.1 states that the amount of dual Peano time needed to pass from $\lambda^N(a)$ to $\lambda^N(r)$ equals the number of edges in $\mathcal{R}_{a,r}$, up to the endpoint convention. Therefore

$$|T_n(r) - T_n(a)| \le \#\mathcal{R}_{a,r} + C.$$

Lemma 3.1 gives, uniformly for $a \le r \le b$,

$$|T_n(r) - T_n(a)| \le (b - a + 1) + \#M^N_{a,b} + \#M^S_{a,b} + C'. \tag{10}$$

Take $a = a_n$ and $b = b_n$. The two pocket terms divided by n are tight by GHS Proposition 5.1. The rescaled dual time $n^{-1}T_n(a_n)$ is tight for rational s by GHS Proposition 5.5. Dividing (10) by n proves (9). The finite-family assertion follows by a union bound. □

Remark 3.3. *This is the only place where adaptivity of the SDP maximizer matters. No raw-strip or adaptive-sector comparison for shortest paths is used.*

4 Countably many marked continuum cells

For $k \in \mathbb{N}_0$, use the exact n-increment block

$$I_{k,n} := (2kn, (2k+1)n]_{\mathbb{Z}}.$$

The unused interval $((2k+1)n, (2k+2)n]_{\mathbb{Z}}$ separates successive selected blocks. Let $\Delta_{k,n}$ be the fresh SDP diameter obtained by running the increments in $I_{k,n}$ afresh. The variables $\Delta_{k,n}$ are i.i.d. copies of Δ_n. Define the block mark

$$B_{k,n} := \mathbf{1}_{\{m_n \le \Delta_{k,n} \le A m_n\}}, \tag{11}$$

where A is from Lemma 2.2. Then

$$p_n := \mathbb{P}[B_{k,n} = 1] \geq c_0 \tag{12}$$

for all large n.

On $\{B_{k,n} = 1\}$, choose a diameter-realizing fresh-block geodesic by a fixed deterministic tie-breaking rule. Lemma 2.1 supplies middle edges $e_{k,n}, f_{k,n}$ in the completed UIBOT with

$$1 - \frac{2}{m_n} \leq H_{k,n} := \frac{d_M^\bullet(e_{k,n}, f_{k,n})}{m_n} \leq A. \tag{13}$$

Their north-going labels lie in $[2kn, (2k+1)n]_{\mathbb{Z}}$. Set

$$X_{k,n} = n^{-1} p(e_{k,n}), \qquad Y_{k,n} = n^{-1} p(f_{k,n}).$$

On failure, set $(X_{k,n}, Y_{k,n}, H_{k,n})$ equal to a fixed cemetery state.

For each fixed k, Lemma 3.2 makes the base coordinates tight. Since $B_{k,n}$ is compact-valued and the successful heights are uniformly bounded, the countable family of marked coordinates is jointly tight with $(\mathcal{W}_n, \Gamma_n, Z_n)$. Explicitly, choose for coordinate k a compact set with failure probability at most $\varepsilon 2^{-k-2}$ and use the union bound; the product of these compact sets is compact in the countable product topology.

Pass to a further subsequence such that $p_n \to p \geq c_0$ and

$$(\mathcal{W}_n, \Gamma_n, Z_n, (B_{k,n}, X_{k,n}, Y_{k,n}, H_{k,n})_{k\geq 0}) \Longrightarrow (\mathcal{W}, \Gamma, Z, (B_k, X_k, Y_k, H_k)_{k\geq 0}). \tag{14}$$

The marginal law of $(\mathcal{W}, \Gamma, Z)$ is unchanged from Input 1.1. The state space in (14) is Polish, so we henceforth use one Skorokhod coupling on which the entire convergence is almost sure.

Proposition 4.1 (Independent marked wedge cells)**.** *The limit has the following properties.*

(i) The marks B_k are i.i.d. Bernoulli(p), with $p \geq c_0$.

(ii) On $\{B_k = 1\}$,

$$((X_k, Y_k), H_k) \in \Gamma, \qquad H_k \in [1, A]. \tag{15}$$

(iii) Let $(\mathcal{S}_{a,b}, \widehat{\eta}_{a,b})$ be the intrinsic curve-decorated quantum surface swept during $[a, b]$, in the notation of GHS Definition 3.11. Then

$$(\mathcal{S}_{2k,2k+1}, \widehat{\eta}_{2k,2k+1}, B_k), \qquad k \geq 0,$$

are i.i.d. marked quantum-surface cells. Each unmarked cell is the first unit of quantum area of the fresh weight-4/3 quantum wedge obtained by rerooting at time $2k$.

Proof. At each n, the vectors consisting of the recentered KMSW walk on $I_{k,n}$ and $B_{k,n}$ are i.i.d. across k, since they are measurable functions of disjoint sets of i.i.d. increments. Independence is preserved under joint weak convergence, and $p_n \to p$, which proves (i).

For (ii), work on the Skorokhod coupling. If $B_k = 1$, then the discrete $\{0, 1\}$-valued convergence implies $B_{k,n} = 1$ eventually. Therefore the prelimit points

$$((X_{k,n}, Y_{k,n}), H_{k,n}) \in \Gamma_n$$

are eventually defined and converge to $((X_k, Y_k), H_k)$, with $H_k \in [1, A]$ by (13). The upper-limit property of Fell convergence puts the limit point in Γ. This holds simultaneously for all k on a countable intersection of probability-one events.

For (iii), GHS Lemma 3.12 states that the intrinsic segment $(\mathcal{S}_{a,b}, \widehat{\eta}_{a,b})$ and the centered Brownian increment $(Z - Z_a)|_{[a,b]}$ almost surely determine one another. Hence adjoining the intrinsic surface to each limiting block does not destroy the independence inherited above. GHS Lemma 3.13 states that after a peanosphere stopping time T, the future intrinsic surface-curve pair is independent of the past and has the law of a $(2 - \gamma^2/2)$-quantum wedge decorated by an independent space-filling $\mathrm{SLE}_{\kappa'}$. Here $\gamma = \sqrt{4/3}$ and $\kappa' = 12$, so the wedge weight is

$$2 - \frac{\gamma^2}{2} = \frac{4}{3}.$$

For deterministic $T = 2k$, the interval $[2k, 2k + 1]$ is the first unit of quantum area of this fresh future wedge. □

Remark 4.2 (Why the cells are marked)**.** *We never assert that B_k, or the limiting directed distance field, is a measurable function of the unmarked LQG/SLE surface. The auxiliary mark is allowed to retain microscopic randomness. The needed independence is stronger and safer: it is inherited from disjoint KMSW increment blocks before the weak limit is taken.*

5 Continuum zero–one amplification

Lemma 5.1 (Zero height on the diagonal)**.** *Almost surely, if $((x, x), h) \in \Gamma$, then $h = 0$.*

Proof. Fix $(x, x) \in \mathcal{W}$. By Fell convergence of the domains, choose $w_n = (u_n, v_n) \in \mathcal{W}_n$ with $w_n \to (x, x)$. Then $u_n \in \mathcal{U}_n$, $u_n \to x$, and $(u_n, u_n) \in \mathcal{W}_n$. Hence

$$\left((u_n, u_n), \frac{1}{m_n}\right) \in \Gamma_n.$$

Since $m_n \to \infty$, these graph points converge to $((x, x), 0)$, which belongs to Γ by Fell convergence. Since Γ is the graph of a function, its fibre over (x, x) contains no other height. □

Proof of Theorem 1.2. By Proposition 4.1(i), $(B_k)_{k \geq 0}$ is an i.i.d. continuum mark sequence with

$$\mathbb{P}[B_k = 1] = p \geq c_0 > 0.$$

Thus the second Borel–Cantelli lemma gives

$$\mathbb{P}[B_k = 1 \text{ for infinitely many } k] = 1.$$

Equivalently, this tail event has probability one by the Kolmogorov zero–one law. This is the continuum regeneration step: after the marked cells have been extracted, no discrete independence estimate is used in the amplification.

For any successful k, Proposition 4.1(ii) gives

$$((X_k, Y_k), H_k) \in \Gamma, \qquad H_k \geq 1.$$

Since $\Gamma = \mathrm{Graph}(D)$,

$$D(X_k, Y_k) = H_k > 0.$$

The pair is comparable because $(X_k, Y_k) \in \mathcal{W}$. If $X_k = Y_k$, then Lemma 5.1 would imply $H_k = 0$, a contradiction. Therefore $X_k \neq Y_k$. Almost surely there are infinitely many such cells, proving the theorem. □

6 Scope of the result

The proof separates cleanly into a discrete local seed and a continuum amplification. The discrete input is unavoidable with the presently constructed objects: there is not yet an independently defined continuum SDP metric whose internal diameter could be normalized and sampled directly. What is continuum here is the regeneration step after the local metric mark has been carried to the scaling limit.

Two stronger assertions are not proved.

(i) The argument does not prove that every median-normalized subsequential closed-set limit is graph-valued; this is exactly Input 1.1.

(ii) It does not prove point separation

$$D(x, y) > 0 \qquad \text{for every distinct comparable } x, y.$$

That would require a positive-cost directed separator for prescribed points, or an equivalent lower crossing estimate in a marked flow-line pocket.

No fixed-ratio regularity of the median sequence is hidden in the proof: every selected discrete cell has exactly n increments and is normalized by the same m_n as the global field.

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